

Ayush Agarwal

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Experience

Docu3C Technologies

AI/ML Intern

- Designed and deployed multi-agent AI systems using MCP (Model Context Protocol), enabling automated tool-calling workflows for financial document processing
- Engineered LLM pipelines with structured outputs (Pydantic, Instructor) serving 3+ document intelligence use cases; reduced manual review time by an estimated 60%
- Architected real-time backend services with FastAPI, SSE, and WebSockets, handling concurrent async requests across multiple AI workflows
- Built full-stack validation interfaces (React, Streamlit) for AI output review; provisioned Azure ML GPU infrastructure, reducing iteration cycles from days to hours
- Enforced PEP8 standards, Ruff linting, and modular architecture across the codebase; managed dependencies with uv

Projects

LexAI – Production AI Platform

Cloud-Native Hybrid RAG

Deployed: lexai-3fd1a.web.app

- Single-handedly built and deployed a full-stack legal intelligence platform indexing 26,274 judgments (1.8GB corpus) with live production traffic
- Combined SQLite FTS5 (BM25), FAISS vector search (MPNet), PageRank graph ranking, and Reciprocal Rank Fusion into a unified retrieval engine
- Achieved 67.8 Recall@10 (+3.6% over baselines) with sub-1.1s query latency on 4Gi RAM / 2 CPU infrastructure
- Integrated Gemini 1.5 Flash for conversational querying; delivered async FastAPI services with REST + WebSocket endpoints
- Deployed React (Vite) frontend on Firebase Hosting with Google Sign-In auth; containerized backend on Google Cloud Run with Docker, autoscaling, and Cloud Build CI/CD

PolarBrief AI – Legal Argument Analyzer

Full-Stack LLM App

- Created document analysis tool that extracts, classifies (Pro/Con/Neutral), and scores legal arguments from unstructured PDFs
- Orchestrated LLMs via LangChain for summarization; combined TF-IDF with LLM outputs for weighted ranking
- Implemented hybrid OCR pipeline (Tesseract + pdfplumber) with Streamlit interface for report generation

Knowledge Graph Pipeline

NLP + Graph ML

- Constructed end-to-end KG extraction pipeline: OCR → chunking → entity-relation extraction → fuzzy resolution → validation
- Visualized canonical knowledge graphs using NetworkX and PyVis for interactive exploration

WESAD Multimodal Emotion Recognition

Machine Learning

- Compared 5 ML models across 7 physiological signal modalities (ECG, EDA, EMG) for emotion classification
- Identified optimal sensor configurations (chest vs wrist) through cross-validation and F1-score evaluation

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| <ul style="list-style-type: none">Languages: Python, TypeScript, JavaScript, HTML/CSSAI/ML: LLMs, RAG, NLP, Multi-Agent Systems, Prompt EngineeringAgentic AI: MCP, Tool Calling, Function Calling, Pydantic, InstructorFrameworks: FastAPI, React, Vite, Node.js, LangChain, Scikit-learnRetrieval: FAISS, BM25, SQLite FTS5, Vector DBs, PageRank, RRF | <ul style="list-style-type: none">Databases: PostgreSQL, Firestore, SQLite, CosmosDBCloud/DevOps: GCP (Run, Build, Storage), Azure, Docker, Firebase, CI/CDAPIs: REST, WebSockets, SSE, AsyncIO, CORS, OAuth 2.0, Google Sign-InTools: Git, GitHub, Tesseract OCR, pdfplumber, NetworkX, PyVisPractices: Agile, PEP8, Ruff, uv, Autoscaling, Health Checks, Testing |
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Education

B. Tech, CSE (Core)

VIT Bhopal University

CGPA: 8.52/10

Sep 2023 – Ongoing

Certifications, Achievements & Additional Information

Certifications	Achievements & Activities
Microsoft Certified: Azure Data Fundamentals (Jun 2025) NPTEL: Intro to Machine Learning (May 2025) Google Cloud: Intro to LLMs (Apr 2024) Google Cloud: Intro to Generative AI (Apr 2024) Google Cloud: Prompt Design in Vertex AI (Jul 2024)	Top 25 team, Mahakumbh Hackathon Solved 125+ DSA problems on LeetCode Event Lead, Linpack Club — 5+ workshops (200+ attendees) University volleyball team (Thunders) — Aavahan fest Languages: English, Hindi Interests: AI agents, Full-stack